

Zomato Restaurant Analytics

1. Project Overview

The Zomato Restaurant Analytics Dashboard is an end-to-end Data Analytics project developed using Python, SQLite, SQL, and Power BI. The project aims to analyze restaurant data and generate meaningful business insights related to restaurant performance, customer preferences, pricing patterns, cuisines, online ordering services, table booking facilities, and restaurant popularity.

Restaurants generate large volumes of customer ratings and operational data. Analyzing this information helps businesses understand customer behavior, identify popular services, optimize pricing strategies, and improve overall customer experience.

This project follows a complete analytics workflow involving data collection, data cleaning, exploratory data analysis (EDA), feature engineering, database creation using SQLite, SQL-based business analysis, and interactive dashboard development using Power BI.

2. Objectives

The primary objectives of this project are:

- Analyze restaurant ratings and customer preferences.
 - Identify the most popular restaurant types and cuisines.
 - Study the impact of online ordering on restaurant ratings.
 - Study the impact of table booking facilities on restaurant ratings.
 - Discover high-density restaurant locations.
 - Analyze restaurant pricing patterns.
 - Develop a custom popularity metric for restaurant evaluation.
 - Build an interactive dashboard for business decision-making.
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3. Technologies Used

Programming and Analysis

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Database

- SQLite
- SQL

Business Intelligence

- Power BI

Version Control

- Git
 - GitHub
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4. Dataset Description

Dataset Information

- Original Records: 7,105
- Records After Cleaning: 6,984

Features Used

- Restaurant Name
 - Restaurant Type
 - Rate (Out of 5)
 - Number of Ratings
 - Average Cost (Two People)
 - Online Order
 - Table Booking
 - Cuisines Type
 - Area
 - Local Address
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5. Data Cleaning and Preprocessing

Several preprocessing steps were performed to improve data quality and ensure accurate analysis.

Cleaning Steps

- Removed unnecessary columns:
 - Unnamed: 0
 - Unnamed: 0.1

- Identified and handled missing values.
- Removed rows containing missing values in critical columns.
- Identified and removed duplicate records.
- Verified dataset consistency.
- Prepared the dataset for analysis and visualization.

Final Dataset

- Total Records: 6,984
 - Total Features: 10
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6. NumPy Statistical Analysis

NumPy was used to perform statistical analysis on restaurant ratings.

Metrics Calculated

- Mean Rating
- Median Rating
- Standard Deviation of Ratings

These statistics provide insights into the overall rating behavior and variability across restaurants.

7. Feature Engineering

A custom metric called Popularity Score was created to measure restaurant influence and popularity.

Formula

Popularity Score = Rating × Number of Ratings

This metric helps identify restaurants that not only receive high ratings but also attract significant customer engagement through reviews.

Benefits of Popularity Score:

- Combines quality and popularity.
 - Prevents highly rated restaurants with very few reviews from dominating rankings.
 - Highlights restaurants with strong customer impact.
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8. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Pandas, NumPy, Matplotlib, and Seaborn.

Restaurant Analysis

- Top Restaurant Types
- Top Rated Restaurant Categories
- Top 10 Most Popular Restaurants
- Restaurant Popularity Analysis

Customer Service Analysis

- Online Ordering Distribution
- Table Booking Distribution
- Average Ratings by Online Order Availability
- Average Ratings by Table Booking Availability

Location Analysis

- Top Areas by Number of Restaurants
- Restaurant Density Analysis

Cuisine Analysis

- Most Popular Cuisines

Rating Analysis

- Rating Distribution
- Mean, Median, and Standard Deviation of Ratings

Cost Analysis

- Cost Distribution
- Cost vs Rating Relationship
- Average Cost by Restaurant Type

9. SQLite Database Integration

To demonstrate database management and SQL analytics skills, the cleaned dataset was stored in a SQLite database.

Database Workflow

Cleaned Dataset (CSV)

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SQLite Database

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SQL Queries



Business Insights



Python Visualizations

Database Table

The SQLite database contains the following table:

- zomato

The database was created and managed using Python's sqlite3 library.

10. SQL Analysis

SQL queries were used to perform business-focused analysis on restaurant data.

Business Questions Answered

1. What is the total number of restaurants?
2. What are the top areas with the highest restaurant density?
3. What percentage of restaurants offer online ordering?
4. What percentage of restaurants offer table booking?
5. How does online ordering affect restaurant ratings?
6. How does table booking affect restaurant ratings?
7. Which restaurant types have the highest average ratings?
8. Which restaurant types have the highest average costs?

SQL Concepts Used

- SELECT
- COUNT()
- AVG()
- GROUP BY
- ORDER BY
- HAVING
- LIMIT

SQL Tools Used

- SQLite
- SQL Queries

- Pandas read_sql_query()
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11. Power BI Dashboard Development

An interactive dashboard was created using Power BI to visualize business insights.

KPI Cards

- Total Restaurants
- Average Rating
- Average Cost for Two People
- Restaurants Offering Online Orders
- Restaurants Offering Table Booking

Visualizations

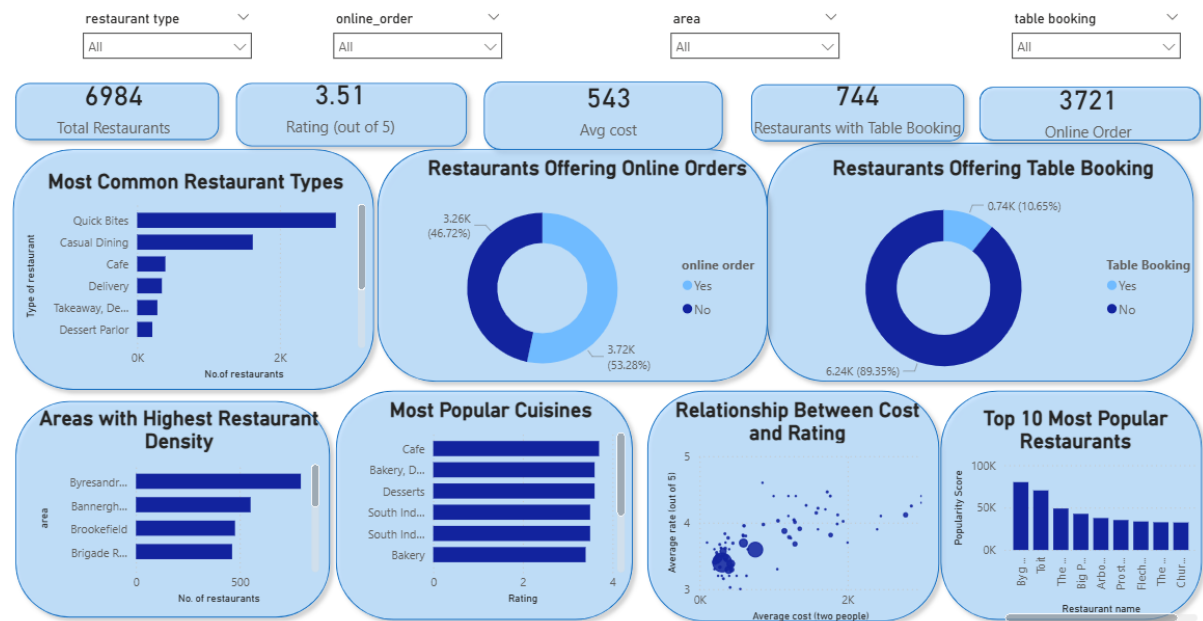
- Most Common Restaurant Types
- Online Order Distribution
- Table Booking Distribution
- Areas with Highest Restaurant Density
- Most Popular Cuisines
- Cost vs Rating Analysis
- Top 10 Most Popular Restaurants (Popularity Score)

Interactive Filters

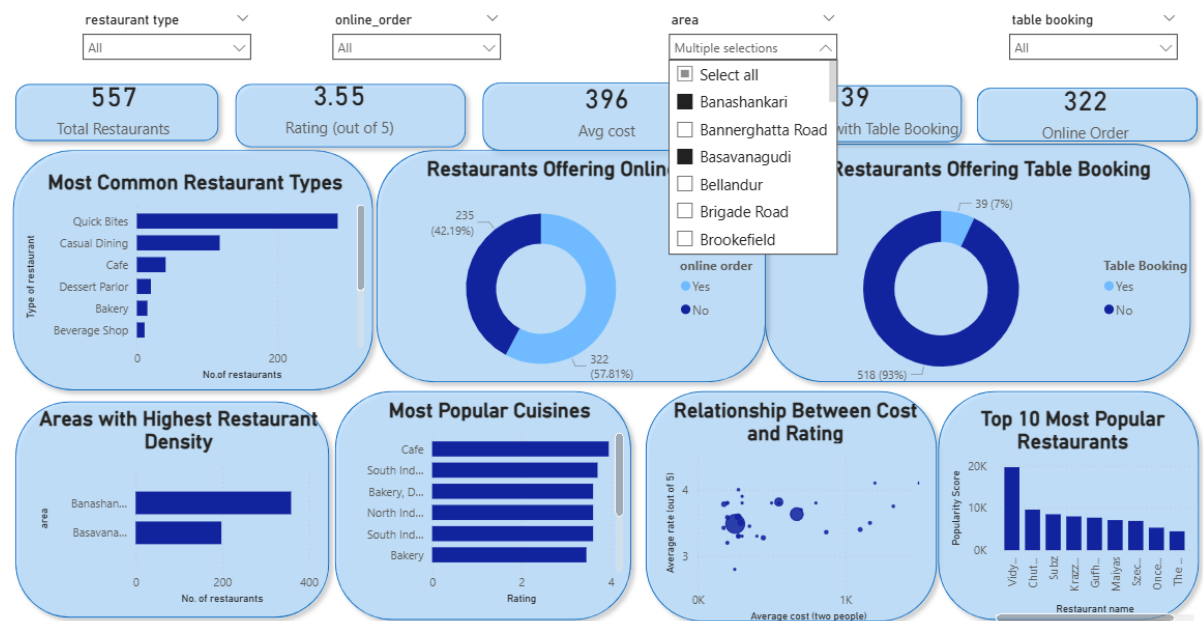
- Area
- Restaurant Type
- Online Order
- Table Booking

The dashboard allows users to interactively explore restaurant trends and business insights.

Zomato Restaurant Analytics Dashboard



Zomato Restaurant Analytics Dashboard



12. Key Findings

- Quick Bites is the most common restaurant type.
- More than 53% of restaurants provide online ordering services.
- Restaurants offering table booking generally have higher average ratings.
- The average restaurant rating is approximately 3.51 out of 5.
- The average cost for two people is approximately ₹543.

- Byresandra–Tavarekere–Madiwala contains the highest concentration of restaurants.
 - Popular cuisines dominate customer preferences across the city.
 - Restaurants with higher ratings and larger review counts achieve higher Popularity Scores.
 - Online ordering services contribute significantly to restaurant visibility and customer engagement.
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13. Dashboard Preview

(Insert Power BI Dashboard Screenshot Here)

The dashboard provides a centralized view of restaurant performance metrics and customer behavior patterns.

14. Project Structure

Zomato-Analytics-Project

```
├── Dataset
├── Notebook
├── Database
├── SQL_Queries
├── PowerBI
├── Reports
├── Screenshots
├── README.md
├── requirements.txt
└── .gitignore
```

15. Future Enhancements

Potential improvements for future versions of the project include:

- Integration with live restaurant datasets.
- Sentiment analysis using customer reviews.
- Machine Learning models for rating prediction.
- Geospatial analysis using restaurant coordinates.
- Real-time dashboard updates.

- Restaurant recommendation systems.
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16. Conclusion

The Zomato Restaurant Analytics Dashboard successfully demonstrates a complete end-to-end data analytics workflow using Python, NumPy, Pandas, SQLite, SQL, and Power BI.

The project transforms raw restaurant data into meaningful business insights through data cleaning, statistical analysis, exploratory data analysis, SQL-based querying, feature engineering, and dashboard reporting.

The findings provide valuable information about customer preferences, restaurant performance, service offerings, pricing strategies, and location-based trends. The interactive dashboard enables data-driven decision-making and highlights practical applications of analytics in the restaurant industry.

This project demonstrates practical skills in data preprocessing, statistical analysis, SQL querying, database management, data visualization, business intelligence reporting, and dashboard development.